Short questions

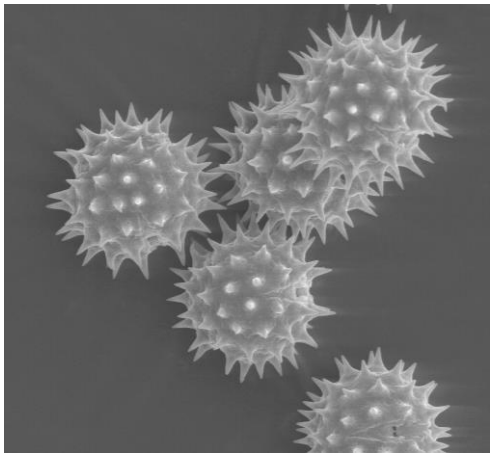
|!|BCESB0140000001|!| ★ **Cross topic**

The following photographs show two structures (X and Y) that are involved in reproduction in humans and plants.



Structures X

Source:http://commons.wikimedia.org/wiki/File:Sperm-20051108.jpg
Credit: Gilberto Santa Rosa



Structures Y

Source:http://commons.wikimedia.org/wiki/File:Helianthus_annuus_pollen_1.jpg

(a) State the sites where structures X and Y are made. (2 marks)

(b) Which kind of cell division contributes to the production of structures X and Y? (1 mark)

(c) According to the photographs, state ***one*** structural feature that helps structures X and Y adapt to reproduction. (2 marks)

##

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(a) Structures X: testis (1)

Structures Y: anther (1)

(b) Meiotic cell division (1)

(c) Structure X has tail for swimming towards ovum. (1)

Structure Y has rough surface to attach to insects / stick to the stigma. (1)

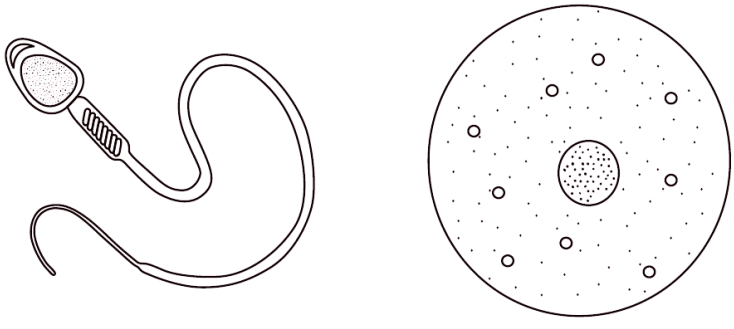
(5 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Analysing data	2

|!|BCESB0140000002|!| ★

The diagrams below show two human sex cells.



cell A cell B

(a) Name cells A and B. (2 marks)

(b) Describe ***two*** features of cell A that are different from cell B. (2 marks)

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(c) Cells A and B are haploid. Explain why it is important that they are haploid.

(2 marks)

##

(a) Cell A: sperm (1)

Cell B: ovum (1)

(b) Any ***two*** of the following: (2)

- Cell A has a tadpole shape while cell B is spherical in shape. (1)
- Cell A can move by beating its tail, while cell B cannot move by itself. (1)
- Cell A has a higher density of mitochondria than cell B. (1)
- Cell A has a smaller amount of food reserves than cell B. (1)

(Or other reasonable answers)

(c) When the nuclei of cells A and B fuse, (1)

the diploid chromosome number is restored in the zygote. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	2

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|!|BCESC0140000003|!|

For each of the structures of the female reproductive system listed in column I, select from column II one phrase that correctly describes it. Put the letter in the space provided.

(4 marks)

Column I		***Column II***
Ovary	(a)_____	A. Produces ova
Oviduct	(b)_____	B. Implantation occurs
Uterus	(c)_____	C. Provides a stable environment for the foetus to develop
Uterine lining	(d)_____	D. Fertilization occurs
		E. Receives the penis during sexual intercourse

##

(a) A (1)
(b) D (1)
(c) C (1)
(d) B (1)

(4 marks)

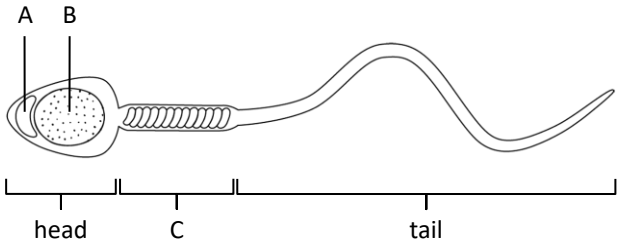
##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESC0140000004|!|

The diagram below shows the structure of a human sperm.



(a) Name the structures labelled A, B and C. (3 marks)

(b) Briefly describe the roles of structures A and C in fertilization. (4 marks)

(c) What sex chromosome does a sperm carry? (1 mark)

##

(a) A: acrosome (1)
B: nucleus (1)
C: midpiece (1)

(b) A contains enzymes (1)
that help the sperm penetrate an ovum. (1)
C is packed with mitochondria (1)
which carry out respiration to supply energy for the movement of the tail. (1)

(c) X or Y chromosome (1)

(8 marks)

##

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	Skill assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	4
(c)	Understanding basic concepts	1

|!|BCESC0140000005|!|

Complete the following paragraph with suitable words. (5 marks)

The cycle of building and shedding the uterine lining is called the (a)_____, which lasts for about 28 days. If fertilization does not occur, from day 1 to day 5, the thickened uterine lining breaks down. Blood, mucus together with the unfertilized ovum are passed out of the body through the (b)_____ as menstrual blood. After menstruation, another follicle becomes mature in the ovary. It produces (c)_____ which causes the uterine lining to repair and thicken. This prepares the uterine lining for receiving an (d)_____. On day 14, the mature follicle releases an ovum into the oviduct. The ruptured follicle becomes the (e)_____ which produces hormones to cause the uterine lining to further thicken. From day 15 to 28, the uterine lining remains thickened. If fertilization does not occur, the uterine lining breaks down and a new cycle begins.

##

(a) menstrual cycle (1)
(b) vagina (1)
(c) oestrogen (1)
(d) embryo (1)
(e) yellow body (1)

(5 marks)

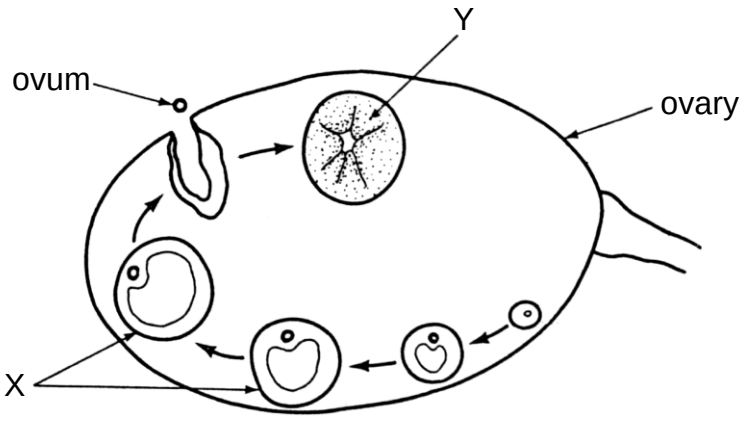
##

	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESC0140000006|!|

The following diagram shows the structures of a human ovary.



(a) What are structures X and Y? (2 marks)

(b) (i) State the change of the uterine lining after the ovum is released from structure X. (1 mark)

(ii) What causes the change stated in (i)? (2 marks)

##

(a) X: follicle (1)

Y: yellow body (1)

(b) (i) The uterine lining becomes further thickened. (1)

(ii) The ruptured follicle becomes the yellow body. (1)

The yellow body produces oestrogen and progesterone which cause the uterine lining to further thicken. (1)

(5 marks)

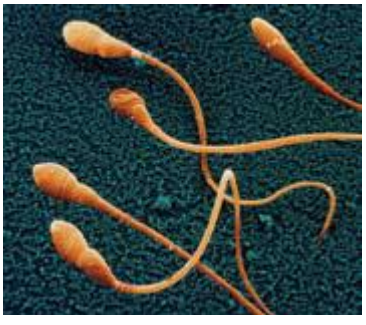
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##

	Skill assessed	Marks
(a)	Analysing data	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2

|!|BCESC0140000007|!|

The following photomicrographs show some human sperms and a human ovum.



Human sperms



A human ovum

(a) Compare the sperm and the ovum in terms of their

(i) sizes, (1 mark)

_____

_____

(ii) appearance, and (1 mark)

_____

_____

(iii) ways of locomotion. (1 mark)

_____

_____

(b) What is the change of an ovum after it is fertilized by a sperm? (1 mark)

_____

##

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(a)

	Ovum	Sperm	
(i) Size	larger	smaller	(1)
(ii) Appearance	spherical	tadpole-like	(1)
(iii) Locomotion	cannot move on their own	can swim	(1)

(b) The jelly coat of an ovum thickens and hardens. (1)

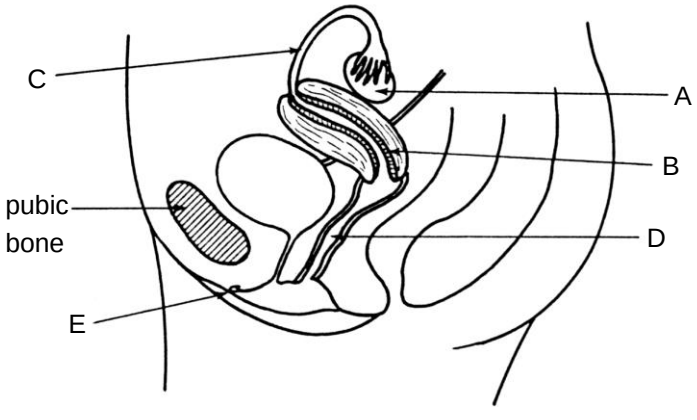
(4 marks)

##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

|!|BCESB0140000008|!| ★

The following diagram shows the structure of the female reproductive system in humans.



(a) Describe the changes of structures A during ovulation. (2 marks)

(b) Describe the changes of structures B during menstruation. (1 mark)

(c) (i) State the labelled structure in which fertilization occurs. (1 mark)

(ii) Describe the process of fertilization. (2 marks)

##

(a) In structure A, the mature follicle grows bigger and moves to the edge of structure A near the oviduct. (1)
The follicle ruptures to release the ovum inside. (1)

(b) In structure B, the thickened uterine lining breaks down during menstruation. (1)

(c) (i) Structure C (1)

(ii) A sperm penetrates the ovum by releasing enzymes to digest a path through the jelly coat of the ovum. (1)
The nucleus of the sperm fuses with the nucleus of the ovum to form a zygote. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	2

|!|BCESB0140000009|!| ★ **STSE**

In vitro fertilization is used by infertile couples to produce their offspring in recent years.

It is a process in which ova and sperms are collected artificially for fertilization outside the body. Several ova are usually collected and fertilized at one time. The fertilized ova are cultured for a period of time and then transferred back into the female's body for implantation.

(a) Suggest ***two*** possible causes of infertility. (2 marks)

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(b) To which part of the female's body would the fertilized ova be transferred for implantation? (1 mark)

(c) Would you suggest an infertile couple to use *in vitro* fertilization to give birth to their children? Explain with ***one*** reason. (1 mark)

##

(a) Any ***two*** of the following: (2)
- Not enough sperms are produced / Sperm count is low. (1)
- Sperms have poor motility. (1)
- The oviducts are blocked. (1)
- No ovulation occurs. (1)

(Or other reasonable answers)

(b) Uterus (1)

(c) Yes

Any ***one*** of the following: (1)
- The man or woman need not to have surgery to treat the infertility problems. (1)
- The woman can give birth of more than one child in one time. (1)

(Or other reasonable answers)

OR

No

Any ***one*** of the following: (1)
- It is costly. (1)
- The success rate is not high. (1)

(Or other reasonable answers)

(4 marks)

##

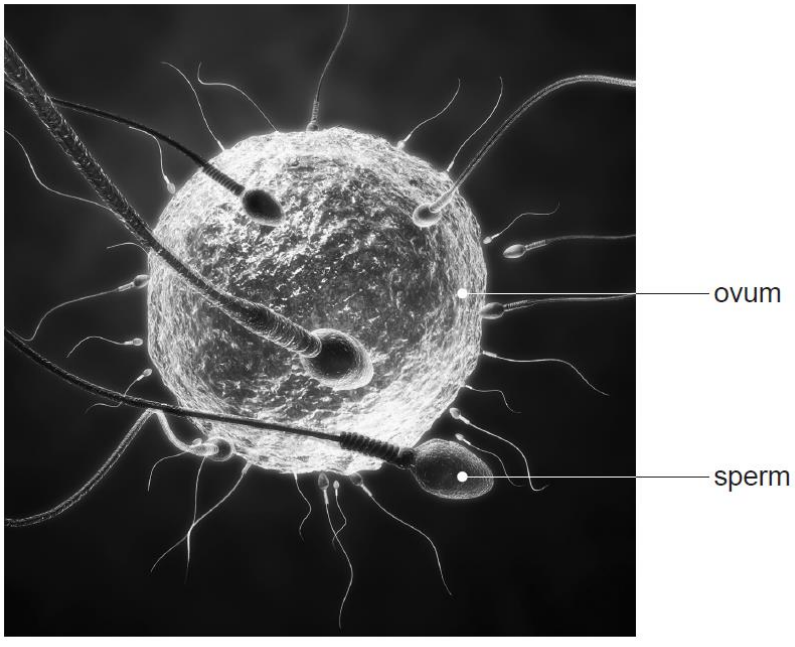
	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Critical thinking	1

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|!|BCESC0140000010|!|

CCEA GCSE Biology Unit 2 Higher Tier 7 Jun 2019 Q5

The photograph shows many sperm and an ovum before fertilization.



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(a) (i) Name the part of a sperm, shown in the photograph, that adapts it to swim towards the ovum. (1 mark)

_____

(ii) Sperm use energy moving towards the ovum. Name the structures found in a sperm which release energy by respiration. (1 mark)

_____

(b) (i) In which structure of the female reproductive system does fertilization normally take place? Describe the process of fertilization. (2 marks)

_____

_____

_____

(ii) Name the cell produced by fertilization. (1 mark)

(iii) Describe what happens to this cell as it moves down the oviduct. (2 marks)

##

(a) (i) Tail (1)
(ii) Mitochondria (1)
(b) (i) Oviduct (1)
The nucleus of a sperm fuses with the nucleus of an ovum. (1)
(ii) Zygote (1)
(iii) The zygote divides by repeated mitotic cell divisions, (1)
forming a ball of cells (embryo). (1)

(7 marks)

##

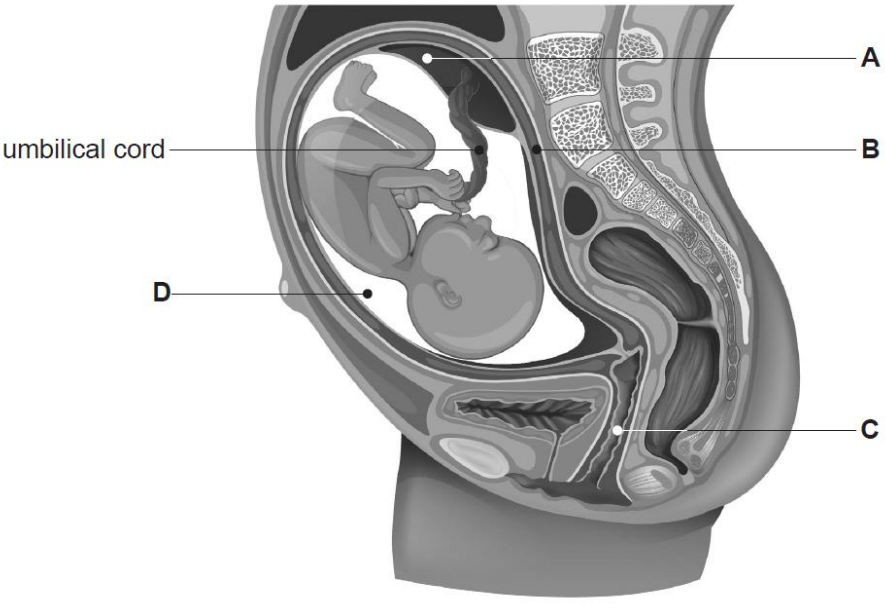
	Skill assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	1
(b)(iii)	Understanding basic concepts	2

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|!|BCESC0140000011|!|

CCEA GCSE Biology Unit 2 Higher Tier 7 Jun 2019 Q4

The diagram shows a foetus in a uterus.



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(a) (i) Name parts A, B and C. (3 marks)

(ii) Name liquid D and describe its function. (2 marks)

(b) Name ***two*** waste substances which pass from the foetus to the mother. (2 marks)

##

(a) (i) A: placenta (1)

B: uterus wall (1)

C: vagina (1)

(ii) Amniotic fluid (1)

Any ***one*** of the following: (1)

- protects the foetus from mechanical shock (1)
- prevents the foetus from drying out (1)
- provides a stable environment in which the foetus can develop (1)
- allows the foetus some freedom of movement (1)
- helps lubricate the birth canal during labour (1)

(b) Urea and carbon dioxide (2)

(7 marks)

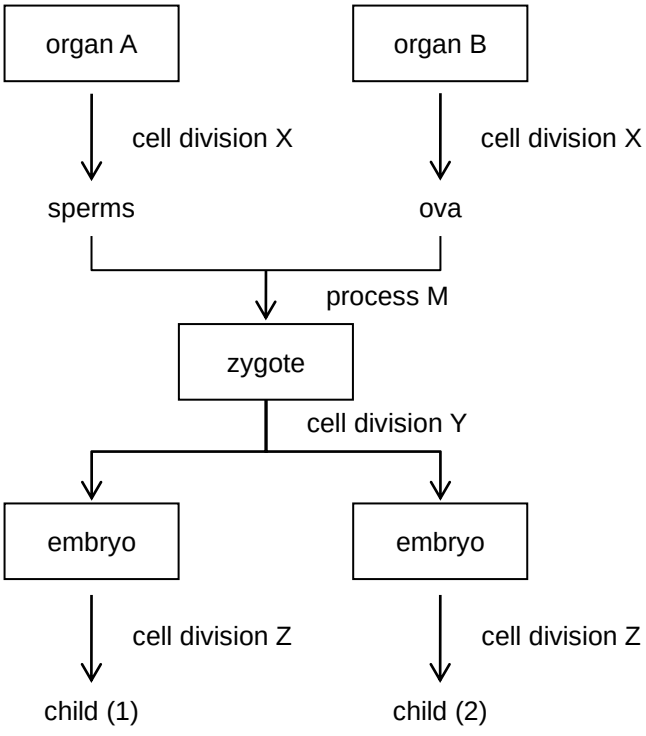
##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	3
(a)(ii)	Understanding basic concepts	2
(b)	Understanding basic concepts	2

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|!|BCESB0140000012|!| ★

The diagram below shows the simplified processes of reproduction in humans.



(a) Name organs A and B. (2 marks)

(b) Name cell divisions X, Y and Z. (3 marks)

(c) Name process M and state where this process takes place. (2 marks)

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(d) Is it possible for child (1) and child (2) to have different sexes? Explain your answer. (2 marks)

##

(a) A: testis (1)
B: ovary (1)
(b) Cell division X: meiotic cell division (1)
Cell division Y: mitotic cell division (1)
Cell division Z: mitotic cell division (1)
(c) Process M: fertilization (1)
It takes place in the oviduct. (1)
(d) Child (1) and child (2) are of the same sex (1)
because they are identical twins / formed from the same zygote. (1)

(9 marks)

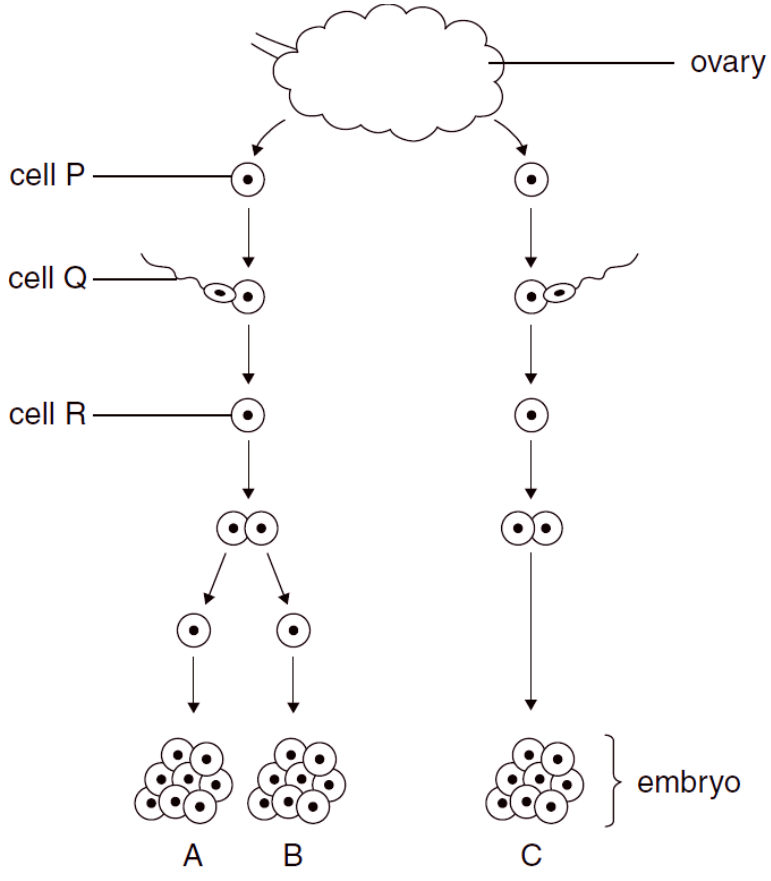
##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	3
(c)	Analysing data	2
(d)	Applying concepts	2

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|!|BCESB0140000013|!| ★

A mother gave birth to triplets: two girls and a boy. The triplets are developed from two ova released from the ovary at the same time. Part of the process is illustrated below:



(a) Identify cells P, Q and R. (3 marks)

(b) Deduce the gender of each of the triplets (A, B or C). Explain your answer. (3 marks)

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##

(a) Cell P: ovum (1)

Cell Q: sperm (1)

Cell R: zygote (1)

(b) Babies A and B are girls, while baby C is a boy. (1)

A and B are developed from the same zygote. They must be genetically identical and are of the same sex. (1)

C is developed from another zygote and is genetically different from A or B. C can be of a different sex. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)	Analysing data	3

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|!|BCESC0140000014|!|

Match the type of twins in column 1 with appropriate descriptions from column 2. Put the appropriate letters in the spaces provided. (4 marks)

Column I		***Column II***
Identical twins	(a)_____	A. They are developed from an ovum fertilized by a sperm and then separated into two parts.
Fraternal twins	(b)_____	B. They can be of different sexes.
		C. They are developed from two ova fertilized by two sperms.
		D. They must be of the same sex.
		E. They are genetically identical.
		F. There are genetic variations between them.

##

(a) A, D, E (1)

(b) B, C, F (1)

(2 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

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